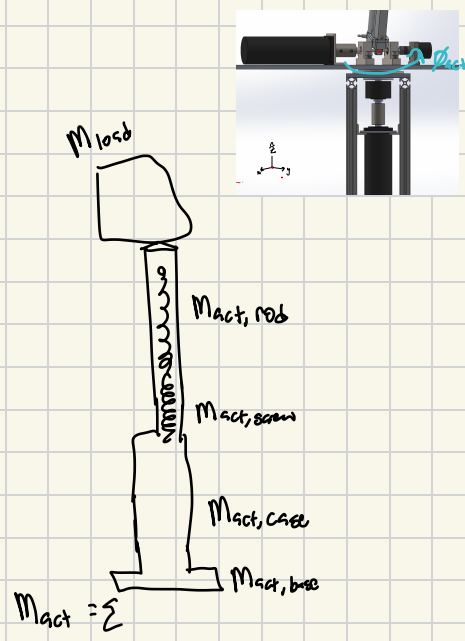
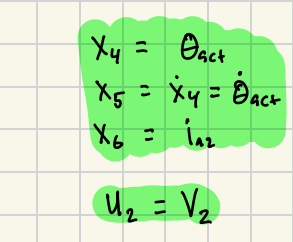




$$u_i = v_i$$

$$\dot{x}_3 = \frac{u_1 - n k_{r1} x_2 - R_{a1} x_3}{L_{a1}}$$



$$\dot{x}_6 = \frac{u_2 - n k_{r2} x_5 - r_{a2} x_6}{L_{a2}}$$

$$\begin{aligned}\dot{x}_2 &= x_8 \\ \dot{x}_8 &= \frac{\rho_s}{2\pi J_{e1,r}} \left[n_3 K_{rs} x_9 - b_{e1,r} \frac{2\pi x_8}{\rho_s} - \frac{(M_{act,rod} + M_{load}) g \cos(x_4) \rho_s}{2\pi} \right] \\ &= \frac{n_3 K_{rs} \rho_s}{2\pi J_{e1,r}} x_9 - \frac{b_{e1,r}}{J_{e1,r}} x_8 + \frac{(M_{act,rod} + M_{load}) g \sin(x_{4,0}) \rho_s^2}{4\pi^2 J_{e1,r}} x_4 \\ \dot{x}_9 &= \frac{u_5 - n_3 K_{rs} \frac{2\pi x_8}{\rho_s} - R_{d3} x_9}{L_{d3}}\end{aligned}$$

$$U_3 = V_3$$